

Yang–Mills Blow-up Theory

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Abstract

The blow-up theory of four-dimensional Yang–Mills connections is a compactness theory with one obstruction: curvature may concentrate at finitely many points. The central result says that after passing to gauges, a bounded-energy sequence of Yang–Mills connections on a compact four-manifold converges smoothly away from finitely many points; at each bad point, one rescales and obtains non-trivial Yang–Mills connections on S^4 , called bubbles; no energy disappears in the neck regions between the base and the bubbles.

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1 The Yang–Mills equation and Basic facts

1.1 The Yang–Mills functional

Let (M^4, g) be a smooth oriented Riemannian four-manifold. Let $E \rightarrow M$ be a real vector bundle of rank r with compact structure group $G \subset \mathrm{SO}(r)$. Denote the Lie algebra by $\mathfrak{g}$, and the associated adjoint bundle by $\mathfrak{g}_E = E \times_G \mathfrak{g}$. A connection D on E induces exterior covariant derivatives

$$d_D : \Omega^k(M, \mathfrak{g}_E) \longrightarrow \Omega^{k+1}(M, \mathfrak{g}_E).$$

Locally, after choosing a trivialization, one writes $D = d + A$, where $A \in \Omega^1(U, \mathfrak{g})$, and the curvature is

$$F_D = dA + A \wedge A \in \Omega^2(U, \mathfrak{g}).$$

Here $A \wedge A$ includes the Lie bracket; in coordinates,

$$(F_D)_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

The Bianchi identity is

$$d_D F_D = 0.$$

A gauge transformation is a section $s \in \Gamma(\text{Aut } E)$, locally a map $s : U \rightarrow G$. It acts by

$$D \mapsto s^* D = s^{-1} D s,$$

so if $D = d + A$, then

$$A \mapsto A^s = s^{-1} d s + s^{-1} A s. \quad (1)$$

The curvature transforms by conjugation:

$$F_{s^* D} = s^{-1} F_D s.$$

Consequently all norms built from the invariant inner product on $\mathfrak{g}$, especially $\int |F_D|^2$, are gauge invariant.

Now we formally introduce the Yang–Mills functional and the Yang–Mills equation. The Yang–Mills functional is defined as

$$\text{YM}(D) = \int_M |F_D|^2 dV_g.$$

If $D_t = D + a_t$ is a variation, then

$$\frac{d}{dt} F_{D_t} = d_{D_t} \dot{a}_t.$$

Therefore

$$\frac{d}{dt} \text{YM}(D_t) = 2 \int_M \langle F_{D_t}, d_{D_t} \dot{a}_t \rangle = 2 \int_M \langle d_{D_t}^* F_{D_t}, \dot{a}_t \rangle.$$

Thus the Euler–Lagrange equation is

$$d_D^* F_D = 0. \quad (2)$$

A connection satisfying (2) is called a *Yang–Mills connection*.

Together with the Bianchi identity, a Yang–Mills connection has curvature satisfying

$$d_D F_D = 0, \quad d_D^* F_D = 0.$$

This resembles the harmonic form equations $dF = d^* F = 0$, except that the operator itself depends on D . This dependence is the source of both the beauty and the difficulty of the theory.

Remark 1.1 (The special role of dimension four). *If g is replaced by $\tilde{g} = e^{2f} g$, then*

$$|F_D|_{\tilde{g}}^2 dV_{\tilde{g}} = e^{-4f} |F_D|_g^2 e^{nf} dV_g.$$

Therefore $\text{YM}(D)$ is conformally invariant exactly when $n = 4$. Also, under the scaling $x \mapsto \lambda x$ on $\mathbb{R}^4$,

$$F \mapsto \lambda^2 F(\lambda x), \quad \int |F|^2 dx \quad \text{is unchanged.}$$

This criticality explains why blow-up produces bubbles on $\mathbb{R}^4$ and, after adding the point at infinity, on S^4 .

1.2 Why gauges are necessary

In a local trivialization, the equation $d_D^* F_D = 0$ has the schematic form

$$d^* dA + A \# \nabla A + A \# A \# A = 0. \quad (3)$$

However, (3) is not elliptic as an equation for A , because the gauge group acts by (1). Infinitesimally, a gauge transformation changes A by

$$A \mapsto A + d_A u.$$

The operator $d^* d$ does not control exact directions du . The standard remedy is to impose a gauge condition, most importantly the *Coulomb gauge*

$$d^* A = 0. \quad (4)$$

Then the system

$$d^* A = 0, \quad dA = F_A - A \wedge A$$

is elliptic for A , with appropriate boundary conditions.

Principle 1.2 (Gauge fixing turns geometry into elliptic analysis). *The Yang–Mills equation becomes useful only after choosing good gauges. Uhlenbeck’s gauge theorem is the basic tool that turns small curvature into analytic control of the connection one-form.*

1.3 The Weitzenbock formula and curvature inequalities

If D is Yang–Mills, then its curvature is harmonic with respect to the covariant Hodge Laplacian. Indeed, the Bianchi identity gives $d_D F_D = 0$, while the Yang–Mills equation gives $d_D^* F_D = 0$. Hence

$$\Delta_D F_D := (d_D d_D^* + d_D^* d_D) F_D = 0.$$

The Weitzenbock formula compares this Hodge Laplacian with the rough Laplacian. Schematically,

$$0 = \Delta_D F_D = \nabla_D^* \nabla_D F_D + \text{Rm} \# F_D + F_D \# F_D. \quad (5)$$

Here $\text{Rm} \# F_D$ denotes the term coming from the Riemannian curvature of the base manifold, while $F_D \# F_D$ is the quadratic term coming from the curvature of the induced connection on the adjoint bundle. On a flat domain in $\mathbb{R}^4$, the background curvature term disappears, and the formula reduces to

$$\nabla_D^* \nabla_D F_D + F_D \# F_D = 0.$$

Taking the inner product with F_D , using Kato’s inequality, and discarding the nonnegative gradient term gives the scalar inequality

$$\Delta |F_D| \geq -C |F_D|^2 \quad (6)$$

in the weak sense. Equivalently, one often works with the slightly less sharp but very useful inequality

$$\Delta |F_D|^2 \geq -C |F_D|^3.$$

This is the analytic starting point for ε -regularity: once the L^2 energy is sufficiently small, the scalar inequality and the mean-value argument convert integral smallness into pointwise curvature control.

Two conformally flat four-manifolds occur repeatedly in the theory: the round sphere S^4 and the cylinder $S^3 \times \mathbb{R}$. The cylinder is conformal to the punctured space $\mathbb{R}^4 \setminus \{0\}$ through the change of variables $x = e^{-t} \theta$, where $\theta \in S^3$ and $t \in \mathbb{R}$. Thus finite energy near an isolated puncture becomes finite energy on a half-cylinder, while neck regions in blow-up analysis become long finite cylinders. Since the Yang–Mills equation and the L^2 energy are conformally invariant in dimension four, these model geometries can be used without changing the basic analytic structure of the problem.

The sphere S^4 has an additional feature: the Weitzenbock formula contains a positive zeroth-order curvature term. This gives a small-energy gap.

Theorem 1.3 (Small-energy gap on S^4). *There exists $\varepsilon_{\text{gap}} > 0$, depending only on the structure group, such that if D is a Yang–Mills connection on a bundle over the round S^4 and*

$$\int_{S^4} |F_D|^2 < \varepsilon_{\text{gap}},$$

then $F_D \equiv 0$.

Proof sketch. By ε -regularity, if ε_{gap} is chosen sufficiently small, then $\|F_D\|_{L^\infty(S^4)}$ is also small. On the round S^4 , the Weitzenbock formula has the schematic form

$$0 = \nabla_D^* \nabla_D F_D + 4F_D + F_D \# F_D.$$

Taking the inner product with F_D and integrating gives

$$0 \geq \int_{S^4} |\nabla_D F_D|^2 + 4|F_D|^2 - C|F_D|^3.$$

If $\|F_D\|_{L^\infty}$ is sufficiently small, then the cubic term is absorbed by the positive quadratic term. Hence

$$\int_{S^4} |\nabla_D F_D|^2 + \int_{S^4} |F_D|^2 = 0,$$

and therefore $F_D \equiv 0$. □

This gap theorem is one of the mechanisms behind energy quantization. In particular, every non-flat bubble on S^4 consumes at least ε_{gap} units of energy. Since the original sequence has uniformly bounded total energy, only finitely many nontrivial bubbles can occur.

2 Uhlenbeck gauge fixing

The following is the analytic engine of the whole theory.

Theorem 2.1 (Uhlenbeck local gauge theorem). *Let E be a G -bundle over the unit ball $B^4 \subset \mathbb{R}^4$, where G is compact, and let D be a smooth connection on E . There exist constants $\varepsilon_U > 0$ and $C < \infty$, depending only on G , such that if*

$$\int_{B^4} |F_D|^2 \leq \varepsilon_U,$$

then there is a trivialization of E over B^4 in which $D = d + A$ and

$$d^* A = 0 \quad \text{in } B^4, \quad \nu \lrcorner A = 0 \quad \text{on } \partial B^4, \quad (7)$$

with

$$\|A\|_{W^{1,2}(B^4)} \leq C \|F_D\|_{L^2(B^4)}. \quad (8)$$

If in addition $F_D \in L^p$ for some $2 < p < 4$, then one also has

$$\|A\|_{W^{1,p}(B^4)} \leq C_p \|F_D\|_{L^p(B^4)}. \quad (9)$$

Proof. Uhlenbeck used the continuity method to prove this local gauge theorem.

Since B^4 is contractible, we first choose an arbitrary background trivialization of E and write $D = d + \tilde{A}$. The point is not to estimate $\tilde{A}$ in this arbitrary trivialization. Instead, one uses the gauge freedom to find a new trivialization in which the connection form satisfies the Coulomb condition and the desired estimates.

For $0 \leq \sigma \leq 1$, define

$$D_\sigma := d + \tilde{A}_\sigma, \quad \tilde{A}_\sigma(x) := \sigma \tilde{A}(\sigma x).$$

Equivalently, D_σ is the pull-back of D by the dilation $x \mapsto \sigma x$. Thus $D_0 = d$ is flat and $D_1 = D$. Its curvature is $F_{D_\sigma}(x) = \sigma^2 F_D(\sigma x)$, and hence, by the scale invariance of the L^2 curvature norm in dimension four,

$$\int_{B^4} |F_{D_\sigma}|^2 = \int_{B_\sigma} |F_D|^2 \leq \int_{B^4} |F_D|^2 \leq \varepsilon_U.$$

Therefore the whole path stays inside the small-energy class. If, in addition, $F_D \in L^p(B^4)$ for some $2 < p < 4$, then the same scaling gives

$$\|F_{D_\sigma}\|_{L^p(B^4)}^p = \sigma^{2p-4} \int_{B_\sigma} |F_D(y)|^p dy \leq \|F_D\|_{L^p(B^4)}^p.$$

Thus the path also stays in the corresponding L^p curvature class.

Let $\mathcal{I} \subset [0, 1]$ be the set of parameters σ for which D_σ admits a gauge in which $D_\sigma = d + A_\sigma$,

$$d^* A_\sigma = 0 \quad \text{in } B^4, \quad \nu \lrcorner A_\sigma = 0 \quad \text{on } \partial B^4,$$

and

$$\|A_\sigma\|_{W^{1,2}(B^4)} \leq C \|F_{D_\sigma}\|_{L^2(B^4)}.$$

If $F_{D_\sigma} \in L^p$ for some $2 < p < 4$, we also require

$$\|A_\sigma\|_{W^{1,p}(B^4)} \leq C_p \|F_{D_\sigma}\|_{L^p(B^4)}.$$

The proof is to show that $\mathcal{I}$ is nonempty, open, and closed. Since $0 \in \mathcal{I}$ trivially, it follows that $\mathcal{I} = [0, 1]$, and in particular $1 \in \mathcal{I}$, which gives the desired gauge for D .

Step 1: Nonemptiness. For $\sigma = 0$, the connection is $D_0 = d$. Taking $A_0 = 0$ gives the Coulomb condition and all estimates. Thus $0 \in \mathcal{I}$.

Step 2: Closedness. Suppose $\sigma_j \in \mathcal{I}$ and $\sigma_j \rightarrow \sigma$. In the fixed background trivialization write $D_{\sigma_j} = d + \tilde{A}_{\sigma_j}$. By definition of $\mathcal{I}$, there are gauge transformations s_j such that

$$A_j = s_j^{-1} ds_j + s_j^{-1} \tilde{A}_{\sigma_j} s_j$$

satisfies the Coulomb condition, the boundary condition, and the uniform estimates. The estimates give weak compactness of A_j in $W^{1,2}$, and in $W^{1,p}$ if the L^p estimate is being used. Moreover, since

$$ds_j = s_j A_j - \tilde{A}_{\sigma_j} s_j,$$

the gauge transformations have weakly convergent subsequences in the corresponding Sobolev spaces. Passing to the limit gives a gauge transformation s and a connection form

$$A = s^{-1} ds + s^{-1} \tilde{A}_\sigma s.$$

The equations $d^* A_j = 0$ and $\nu \lrcorner A_j = 0$ pass to the limit weakly, and the estimates pass to the limit by weak lower semicontinuity. Hence $\sigma \in \mathcal{I}$, so $\mathcal{I}$ is closed.

Step 3: Openness. Assume $\sigma \in \mathcal{I}$. After applying the corresponding gauge, we may write $D_\sigma = d + A$ with

$$d^* A = 0, \quad \nu \lrcorner A = 0.$$

Let $d + A + \lambda$ be a nearby connection form. We look for a small gauge transformation $s = e^u$ such that the transformed connection is again in Coulomb gauge with the same boundary condition. Thus we solve

$$\begin{aligned} d^* (e^{-u} de^u + e^{-u} (A + \lambda) e^u) &= 0 \quad \text{in } B^4, \\ \nu \lrcorner (e^{-u} de^u + e^{-u} (A + \lambda) e^u) &= 0 \quad \text{on } \partial B^4. \end{aligned}$$

Linearizing at $u = 0$ and $\lambda = 0$ gives

$$u \mapsto (d^*(du + [A, u]), \nu \lrcorner (du + [A, u])).$$

Its principal part is the Neumann Laplacian,

$$u \mapsto (d^* du, \partial_\nu u),$$

which is invertible after imposing the normalization $\int_{B^4} u = 0$. The terms involving A are lower order. Since the $W^{1,2}$ norm of A is small when ε_U is chosen small, this linearized operator remains invertible. The implicit function theorem then gives the required gauge for all nearby connections. Hence $\mathcal{I}$ is open.

Step 4: A priori estimate. It remains to proof (8) and (9).

On the ball, the Hodge elliptic estimate for one-forms with normal boundary condition gives

$$\|A\|_{W^{1,p}(B^4)} \leq C_p (\|dA\|_{L^p(B^4)} + \|d^* A\|_{L^p(B^4)}).$$

Here the possible harmonic one-form kernel is absent on B^4 . If A is in Coulomb gauge, then $d^* A = 0$. Using the curvature identity $F_A = dA + A \wedge A$, we obtain

$$\|A\|_{W^{1,p}(B^4)} \leq C_p \|F_A\|_{L^p(B^4)} + C_p \|A \wedge A\|_{L^p(B^4)}. \quad (10)$$

For $2 \leq p < 4$, set $q = 4p/(4-p)$, so that $1/p = 1/4 + 1/q$. Hölder's inequality and the Sobolev embedding $W^{1,p}(B^4) \hookrightarrow L^q(B^4)$ give

$$\|A \wedge A\|_{L^p(B^4)} \leq C \|A\|_{L^4(B^4)} \|A\|_{W^{1,p}(B^4)}. \quad (11)$$

Combining (10) and (11) yields

$$\|A\|_{W^{1,p}(B^4)} \leq C_p \|F_A\|_{L^p(B^4)} + C_p \|A\|_{L^4(B^4)} \|A\|_{W^{1,p}(B^4)}. \quad (12)$$

Now fix p . Choose $\delta_p > 0$ so small that $C_p \delta_p \leq 1/2$. Whenever the Coulomb connection form satisfies $\|A\|_{L^4(B^4)} \leq \delta_p$, (12) gives

$$\|A\|_{W^{1,p}(B^4)} \leq 2C_p \|F_A\|_{L^p(B^4)}.$$

It remains to explain where the smallness of $\|A\|_{L^4}$ comes from. In the continuity argument we follow the small Coulomb-gauge branch starting from the flat connection, where $A = 0$. At a parameter value already in $\mathcal{I}$, the defining estimate gives

$$\|A\|_{W^{1,2}(B^4)} \leq C\|F_A\|_{L^2(B^4)}.$$

Since the whole dilation path satisfies $\|F_A\|_{L^2(B^4)} \leq \varepsilon_U^{1/2}$, Sobolev embedding gives

$$\|A\|_{L^4(B^4)} \leq C\|A\|_{W^{1,2}(B^4)} \leq C\varepsilon_U^{1/2}.$$

Thus, after choosing ε_U sufficiently small, the known Coulomb form has small L^4 norm. In the openness step, the implicit function theorem produces the Coulomb gauge for a nearby connection close to this known small Coulomb form, so the new connection form still satisfies the same L^4 smallness. Applying the absorption argument first with $p = 2$ gives

$$\|A\|_{W^{1,2}(B^4)} \leq C\|F_A\|_{L^2(B^4)}.$$

Then the resulting $W^{1,2}$ estimate again gives the required L^4 smallness, and applying the same absorption argument for any fixed $2 < p < 4$ gives

$$\|A\|_{W^{1,p}(B^4)} \leq C_p\|F_A\|_{L^p(B^4)}.$$

□

Remark 2.2. *The theorem has three important meanings.*

- *First, after choosing a good gauge, small curvature controls the local connection form. Without a gauge condition, the curvature does not control the full first-order behavior of the connection form. A gauge transformation changes A by $A \mapsto s^{-1}ds + s^{-1}As$, and the pure gauge term $s^{-1}ds$ can be large without representing genuine curvature. The Coulomb condition removes this freedom. In that gauge,*

$$d^*A = 0, \quad dA = F_A - A \wedge A,$$

so elliptic estimates apply to A .

- *Second, the hypothesis is scale invariant. In dimension four,*

$$\int_{B_r(x)} |F_D|^2$$

is unchanged under rescaling the ball to unit size. Hence the theorem can be applied uniformly on all small balls where the energy is less than ε_U .

- *Third, the theorem can also be viewed as a local slice theorem for the gauge action. A gauge transformation moves a connection along its gauge orbit. The Coulomb condition selects a preferred representative on this orbit: infinitesimally, the gauge directions are generated by forms of the type $d\xi$, and the condition $d^*A = 0$ says that A is L^2 -orthogonal to these directions, up to the lower-order terms coming from A itself.*

Remark 2.3. *The boundary condition $\nu \lrcorner A = 0$ is the natural Neumann-type condition. Uhlenbeck's original notation on the Euclidean unit ball is $x \cdot A = 0$ on S^3 .*

Remark 2.4. *Uhlenbeck's removable singularity paper also constructs Hodge gauges under small L^∞ -curvature hypotheses on balls, spheres, and annuli. The core geometric idea is to first use an exponential gauge, estimate A by the curvature, and then solve the nonlinear Coulomb equation by an implicit function theorem.*

On annuli, one needs a version compatible with both boundary spheres. This leads to broken Hodge gauges: the connection is in Coulomb gauge on each dyadic annulus, and the tangential parts match across the common boundary spheres. These gauges are crucial in Uhlenbeck's original removable singularity proof.

3 Compactness: first without bubbling, then with bubbling

3.1 Compactness when $p > 2$

If one has a curvature bound stronger than energy, bubbling is excluded. Uhlenbeck's global compactness theorem says:

Theorem 3.1 (Uhlenbeck compactness for L^p -bounded curvature). *Let M^4 be compact and let $p > 2$. Suppose D_i is a sequence of connections on a fixed bundle $E \rightarrow M$ satisfying*

$$\int_M |F_{D_i}|^p \leq C.$$

Then, after passing to a subsequence, there are gauge transformations s_i such that $s_i^ D_i$ converges weakly in $W^{1,p}$ to a $W^{1,p}$ connection D_∞ .*

Proof. By Hölder's inequality, for every geodesic ball $B_r(x)$ one has

$$\begin{aligned} \int_{B_r(x)} |F_{D_i}|^2 &\leq \left(\int_{B_r(x)} |F_{D_i}|^p \right)^{2/p} \text{Vol}(B_r(x))^{1-2/p} \\ &\leq C^{2/p} \text{Vol}(B_r(x))^{1-2/p}. \end{aligned}$$

Since $p > 2$, we may choose $r > 0$ so small that this quantity is below the Uhlenbeck small-energy threshold, uniformly in i and in x .

Choose a finite cover by such balls, slightly enlarged when necessary. On each ball the local Coulomb gauge Theorem 2.1 gives gauges in which $D_i = d + A_{i,\alpha}$ and the forms $A_{i,\alpha}$ are uniformly bounded in $W^{1,p}$. After passing to a subsequence, diagonalizing over the finite cover, we get

$$A_{i,\alpha} \rightharpoonup A_{\infty,\alpha} \quad \text{weakly in } W^{1,p}.$$

On overlaps, the local gauges are related by transition functions $g_{i,\alpha\beta}$. Comparing the two expressions for the same connection gives an equation of the form

$$dg_{i,\alpha\beta} = g_{i,\alpha\beta} A_{i,\alpha} - A_{i,\beta} g_{i,\alpha\beta}, \tag{13}$$

up to the chosen convention for gauge action. Differentiating (13) and using the uniform $W^{1,p}$ bounds for the local connection forms gives uniform $W^{2,p}$ bounds for $g_{i,\alpha\beta}$ on smaller overlaps. Since $p > 2$ in dimension four, $W^{2,p}$ embeds compactly into C^0 . Therefore, after passing to a further subsequence, the transition functions converge strongly in C^0 and weakly in $W^{2,p}$.

The limiting transition functions define a $W^{2,p}$ bundle, and the limiting local forms $A_{\infty,\alpha}$ are compatible with these transition functions; hence they define a global $W^{1,p}$ connection on the limiting bundle. The strong C^0 convergence of the transition functions, together with Uhlenbeck's bundle patching argument, identifies this limiting bundle with the original fixed bundle E after suitable global gauge transformations. In these global gauges, the sequence $s_i^* D_i$ converges weakly in $W^{1,p}$ to a $W^{1,p}$ connection D_∞ on E . $\square$

Remark 3.2. *The borderline energy case $p = 2$ is different: L^2 curvature can concentrate at points. Blow-up theory is precisely the analysis of this concentration.*

3.2 ε -regularity theorem

The next result is the analytic input which converts small four-dimensional energy into pointwise curvature control. It is the reason why, away from energy concentration, a sequence of Yang–Mills connections converges smoothly after gauge transformations.

Theorem 3.3 (ε -regularity). *There exist constants $\varepsilon_{\text{reg}} > 0$ and $C_k < \infty$ with the following property. Let D be a smooth Yang–Mills connection on a geodesic ball $B_{2r}(x) \subset M^4$. If*

$$\int_{B_{2r}(x)} |F_D|^2 \leq \varepsilon_{\text{reg}},$$

then, for every $k \geq 0$,

$$\sup_{B_r(x)} |\nabla_D^k F_D| \leq C_k r^{-2-k} \left(\int_{B_{2r}(x)} |F_D|^2 \right)^{1/2}. \quad (14)$$

In particular,

$$\sup_{B_r(x)} |F_D| \leq C r^{-2} \|F_D\|_{L^2(B_{2r}(x))}.$$

Proof. By scaling and working in normal coordinates, it is enough to prove the estimate on $B_2 \subset \mathbb{R}^4$. On a geodesic ball the additional background curvature terms are lower order and are absorbed into the constants. Set $E := \int_{B_2} |F_D|^2$.

We first prove the curvature bound. Consider

$$Q := \max_{0 \leq \rho \leq 3/2} \left((3/2 - \rho)^4 \sup_{B_\rho} |F_D|^2 \right).$$

Choose ρ_0 and $x_0 \in B_{\rho_0}$ so that $Q = (3/2 - \rho_0)^4 |F_D(x_0)|^2$. Write $e_0 := |F_D(x_0)|^2$ and $\lambda := e_0^{-1/4}$. We claim that $Q \leq 16$ if ε_{reg} is sufficiently small.

If the claim is false, then $\lambda \leq \frac{1}{2}(3/2 - \rho_0)$, and the maximal choice of x_0 gives

$$|F_D(y)|^2 \leq 16e_0 \quad \text{for } y \in B_\lambda(x_0).$$

After rescaling $B_\lambda(x_0)$ to B_1 , the rescaled Yang–Mills connection $\tilde{D}$ satisfies

$$|\tilde{F}(0)| = 1, \quad \sup_{B_1} |\tilde{F}|^2 \leq 16, \quad \int_{B_1} |\tilde{F}|^2 \leq E.$$

The Weitzenböck formula for a Yang–Mills connection gives the scalar inequality

$$\Delta |\tilde{F}|^2 \geq -C |\tilde{F}|^3 \geq -C |\tilde{F}|^2.$$

Hence the mean-value inequality implies

$$1 = |\tilde{F}(0)|^2 \leq C \int_{B_1} |\tilde{F}|^2 \leq CE,$$

which is impossible for $E \leq \varepsilon_{\text{reg}}$ with ε_{reg} sufficiently small. Thus $Q \leq 16$.

In particular, $\sup_{B_{5/4}} |F_D| \leq C$. Returning to the same Weitzenböck inequality, we get on $B_{5/4}$

$$\Delta |F_D|^2 \geq -C |F_D|^2.$$

Another application of the mean-value inequality gives

$$\sup_{B_1} |F_D|^2 \leq C \int_{B_{5/4}} |F_D|^2 \leq CE.$$

Therefore

$$\sup_{B_1} |F_D| \leq C \left(\int_{B_2} |F_D|^2 \right)^{1/2}.$$

It remains to obtain the higher derivative estimates. After shrinking the ball slightly, Uhlenbeck's local gauge theorem gives a Coulomb gauge $D = d + A$ on, say, $B_{3/2}$ with $d^* A = 0$ and, for some fixed $q \in (2, 4)$,

$$\|A\|_{W^{1,q}(B_{3/2})} \leq C \|F_D\|_{L^q(B_{3/2})}.$$

Using the already proved L^∞ bound and the L^2 energy bound, $\|F_D\|_{L^q(B_{3/2})} \leq CE^{1/2}$ after fixing, for instance, $q = 3$. Hence $\|A\|_{W^{1,q}(B_{3/2})} \leq CE^{1/2}$.

In this Coulomb gauge the Yang–Mills equation is an elliptic system

$$\Delta A = A \# \nabla A + A \# A \# A,$$

Interior elliptic estimates and bootstrapping then give, for every $k \geq 0$,

$$\sup_{B_1} |\nabla_D^k F_D| \leq C_k E^{1/2}.$$

Here the smallness of E is used to absorb the nonlinear terms at the first step of the bootstrap; the later steps are standard interior estimates.

Rescaling back to $B_{2r}(x)$ gives (14). This proves the theorem. $\square$

Remark 3.4. *The role of Theorem 3.3 is different from that of the local Uhlenbeck gauge theorem. The gauge theorem gives a good trivialisation under small energy; the ε -regularity theorem uses the Yang–Mills equation to turn that small energy into pointwise and higher derivative curvature bounds. In the compactness argument below, both inputs are used together.*

3.3 Energy concentration set

Let D_i be smooth Yang–Mills connections on a compact four-manifold M^4 satisfying

$$\int_M |F_{D_i}|^2 \leq \Lambda.$$

After passing to a subsequence, the energy measures $|F_{D_i}|^2 d\mu_g$ converge weakly as Radon measures to a finite measure μ on M . Let $\varepsilon_{\text{gauge}}$ be the small-energy constant in Uhlenbeck’s local gauge theorem, and let ε_{reg} be the constant in Theorem 3.3. Fix once and for all $\varepsilon_* < \min\{\varepsilon_{\text{gauge}}, \varepsilon_{\text{reg}}\}$, and define

$$\mathcal{S} = \{x \in M : \mu(\{x\}) \geq \varepsilon_*\}.$$

Then $\mathcal{S}$ is finite and

$$\#\mathcal{S} \leq \Lambda/\varepsilon_*.$$

Indeed, each point of $\mathcal{S}$ carries at least ε_* units of limiting energy, while $\mu(M) \leq \Lambda$.

The definition of $\mathcal{S}$ is designed so that energy is uniformly small away from it. If $x \notin \mathcal{S}$, then $\mu(\{x\}) < \varepsilon_*$. Hence, after choosing $r > 0$ sufficiently small and avoiding at most countably many radii, one has $\mu(\overline{B_{2r}(x)}) < \varepsilon_*$. By weak convergence of measures, it follows that, for all sufficiently large i ,

$$\int_{B_{2r}(x)} |F_{D_i}|^2 < \varepsilon_*.$$

Thus, on balls away from $\mathcal{S}$, the energy is eventually below both the Uhlenbeck gauge threshold and the ε -regularity threshold.

Consequently, on each compact subset $K \Subset M \setminus \mathcal{S}$, one can cover K by finitely many such balls. The ε -regularity theorem gives uniform curvature bounds, and the local Uhlenbeck gauge theorem gives Coulomb gauges. In these gauges the Yang–Mills equation is elliptic, so elliptic bootstrapping gives uniform bounds for the local connection forms in all higher norms. A diagonal argument then gives smooth convergence on compact subsets of $M \setminus \mathcal{S}$.

Theorem 3.5 (Compactness away from concentration points). *Let D_i be smooth Yang–Mills connections on a compact four-manifold M^4 with $\text{YM}(D_i) \leq \Lambda$. After passing to a subsequence, there exist a finite set $\mathcal{S} \subset M$, a smooth Yang–Mills connection D_∞ on a bundle $E_\infty \rightarrow M \setminus \mathcal{S}$, and gauges s_i over compact subsets of $M \setminus \mathcal{S}$, such that*

$$s_i^* D_i \longrightarrow D_\infty$$

smoothly on compact subsets of $M \setminus \mathcal{S}$.

The theorem gives a smooth limiting Yang–Mills connection only on $M \setminus \mathcal{S}$. Thus the compactness statement naturally leaves one local question unresolved: what happens near the points of $\mathcal{S}$? If $x \in \mathcal{S}$ and $B_r(x)$ contains no other point of $\mathcal{S}$, then D_∞ is a smooth Yang–Mills connection on the punctured ball $B_r(x) \setminus \{x\}$. Moreover its energy on this punctured ball is finite, because it is controlled by the limiting energy measure. Therefore the remaining local problem is precisely the finite-energy isolated-singularity problem.

4 Removable singularities

Now we introduce the removable singularity theorem. This theorem is one of the foundations of Yang–Mills compactness. It says that finite-energy point singularities are not genuine singularities; they are gauge artifacts.

Theorem 4.1 (Uhlenbeck removable singularity theorem). *Let D be a smooth Yang–Mills connection on a G -bundle over $B^4 \setminus \{0\}$. If*

$$\int_{B^4} |F_D|^2 < \infty,$$

then there is a gauge in which the bundle extends smoothly over 0 and D extends to a smooth Yang–Mills connection on B^4 .

Uhlenbeck’s original proof. Uhlenbeck’s proof proceeds in three conceptual steps.

Step 1: reduce to small energy near the puncture. Since $F_D \in L^2$, for every $\varepsilon > 0$ there is $\rho > 0$ such that

$$\int_{B_\rho} |F_D|^2 < \varepsilon^2.$$

By conformal/scaling invariance, one rescales this punctured ball to a standard ball and assumes the energy is small.

Step 2: construct a broken Hodge gauge on dyadic annuli. Let

$$U_\ell = \{2^{-\ell-1} \leq |x| \leq 2^{-\ell}\}, \quad S_\ell = \{|x| = 2^{-\ell}\}.$$

On each U_ℓ , Uhlenbeck constructs a gauge with

$$d^* A(\ell) = 0,$$

matching tangential components across the boundary spheres and satisfying suitable boundary conditions. This is the broken Hodge gauge. It is not a single smooth Coulomb gauge across all annuli, but it is coherent enough for integration by parts.

Step 3: derive an energy decay inequality. Set

$$e(r) = \int_{B_r} |F_D|^2.$$

Using the Yang–Mills equation, the broken Hodge gauge, and boundary cancellation between adjacent annuli, Uhlenbeck obtains a differential inequality of the form

$$(1 - \omega e(2r)^{1/2})e(r) \leq \frac{r}{4}e'(r), \tag{15}$$

for small r . Since $e(2r)$ is small, (15) implies improved decay:

$$e(r) \lesssim r^{4\alpha}$$

for some $\alpha > 0$. Combining this with ε -regularity yields bounded curvature near the puncture.

Once $|F_D|$ is bounded, Uhlenbeck applies the local gauge theorem on the whole ball to find a genuine Coulomb gauge with $A \in W^{1,p}$ for some $p > 2$. The weak Yang–Mills equation is then elliptic in Coulomb gauge, so elliptic regularity gives smoothness. $\square$

Cylinder proof. We follow the conformal-cylinder proof, as presented in the lecture notes following the appendix of Freed–Uhlenbeck. Since the curvature has finite L^2 energy, after shrinking the ball and rescaling, we may assume

$$\int_{B^4 \setminus \{0\}} |F_D|^2 < \varepsilon_0,$$

where ε_0 is chosen sufficiently small. Identify the punctured ball with the half-cylinder by

$$B^4 \setminus \{0\} \cong S^3 \times [0, \infty), \quad x = e^{-t}\theta.$$

In dimension four the Yang–Mills equation and the L^2 energy are conformally invariant, so the pulled-back connection is still Yang–Mills and has small total energy on the cylinder.

By ε -regularity, the curvature is uniformly small on the end. On $S^3 \times [0, \infty)$ the Weitzenböck formula has the form

$$\nabla_D^* \nabla_D F_D + \mathcal{R}^\nabla(F_D) + 2F_D = 0,$$

where the positive zeroth-order term comes from the scalar curvature of $S^3 \times \mathbb{R}$. Since $|F_D|$ is small, the quadratic term $\mathcal{R}^\nabla(F_D)$ is dominated by the positive linear term. Thus, for some $c > 0$,

$$\Delta|F_D|^2 \geq c|F_D|^2$$

on the end of the cylinder. A maximum-principle comparison on finite cylinders $S^3 \times [0, R]$, followed by letting $R \rightarrow \infty$, gives exponential decay:

$$|F_D|(\theta, t) \leq Ce^{-\alpha t}.$$

By applying ε -regularity again, the same type of decay holds for the covariant derivatives of F_D on slightly shorter subcylinders.

It remains to use the exponential decay to remove the possible topological singularity. Fix $p > 2$ sufficiently close to 2. For each large integer k , we first construct a trivialization on the slice $S^3 \times \{3k\}$. Cover S^3 by two slightly enlarged hemispheres U_+ and U_- . On each hemisphere, Uhlenbeck's local gauge theorem gives a gauge in which the connection form satisfies

$$\|A_\pm\|_{W^{1,p}(U_\pm \times \{3k\})} \leq Ce^{-\alpha'k}.$$

On the overlap, the transition function g satisfies

$$A_- = g^{-1}dg + g^{-1}A_+g.$$

Hence dg is exponentially small. After multiplying one of the two gauges by a constant element of G , the transition function is close to the identity. A fixed cutoff then patches the two local gauges into a global trivialization of $E|_{S^3 \times \{3k\}}$, still with

$$\|A\|_{W^{1,p}(S^3 \times \{3k\})} \leq Ce^{-\alpha'k}.$$

We next extend this trivialization to the slab $\Omega_k := S^3 \times [3k-2, 3k+2]$ by parallel transport in the t -direction. In this gauge, $A_t := A(\partial_t) = 0$. Therefore the tangential part satisfies $\partial_t A_\theta = F_{t\theta}$ (where $A_\theta := A(\theta)$). Together with the curvature identity for $F_{\theta\theta}$ and the exponential decay of F_D and its covariant derivatives on the cylinder, this gives

$$\|A_k\|_{W^{1,p}(\Omega_k)} \leq Ce^{-\alpha'k}.$$

On the overlaps of consecutive slabs, the corresponding transition functions again have exponentially small derivative, and hence are close to constants. Adjusting the gauges by these constants and then patching inductively, we obtain one trivialization on the cylindrical end $S^3 \times [T, \infty)$ such that

$$\|A\|_{W^{1,p}(S^3 \times [k, k+1])} \leq Ce^{-\alpha''k}$$

for some $\alpha'' > 0$.

Finally return to the punctured ball by $r = e^{-t}$. Since $p > 2$ is chosen sufficiently close to 2, the above exponential decay is strong enough to imply that the connection form is in $W^{1,p}$ across $r = 0$. Thus the bundle extends across the origin as a Sobolev G -bundle, and D extends as a $W^{1,p}$ Yang–Mills connection. Applying the local Coulomb gauge theorem on the ball, we put the extended connection in Coulomb gauge. In this gauge the Yang–Mills equation is an elliptic system for the connection form, so standard elliptic bootstrapping upgrades the Sobolev extension to a smooth Yang–Mills connection. This proves the removable singularity theorem. $\square$

Remark 4.2. A useful corollary is obtained by inversion. If D is Yang–Mills on an exterior region $\{|x| \geq R\} \subset \mathbb{R}^4$ and has finite energy, then after the conformal inversion

$$x \mapsto \frac{x}{|x|^2},$$

the problem becomes a punctured-ball problem. Thus D extends smoothly over the point at infinity. Equivalently, a finite-energy Yang–Mills connection on $\mathbb{R}^4$ extends to a smooth Yang–Mills connection on S^4 .

5 Blow-up and bubbles

5.1 The basic blow-up picture

Let D_i be smooth Yang–Mills connections on a compact four-manifold M with uniformly bounded energy, i.e. $\text{YM}(D_i) \leq \Lambda$. After passing to a subsequence, the connections converge smoothly away from a finite concentration set $\mathcal{S}$. Fix a point $x_0 \in \mathcal{S}$. Blow-up analysis studies the geometry seen at smaller and smaller scales around x_0 .

Choose points $x_i \rightarrow x_0$ and scales $\lambda_i \downarrow 0$. In normal coordinates, define $\phi_i(y) = \exp_{x_i}(\lambda_i y)$. The pulled-back connections $\tilde{D}_i = \phi_i^* D_i$ are defined on larger and larger balls in $\mathbb{R}^4$. More precisely, they are Yang–Mills with respect to the rescaled metrics $g_i := \lambda_i^{-2} \phi_i^* g$, and $g_i \rightarrow g_{\text{euc}}$ smoothly on compact subsets of $\mathbb{R}^4$. By the conformal invariance of the four-dimensional Yang–Mills equation and of the L^2 curvature energy, the energy on corresponding regions is unchanged.

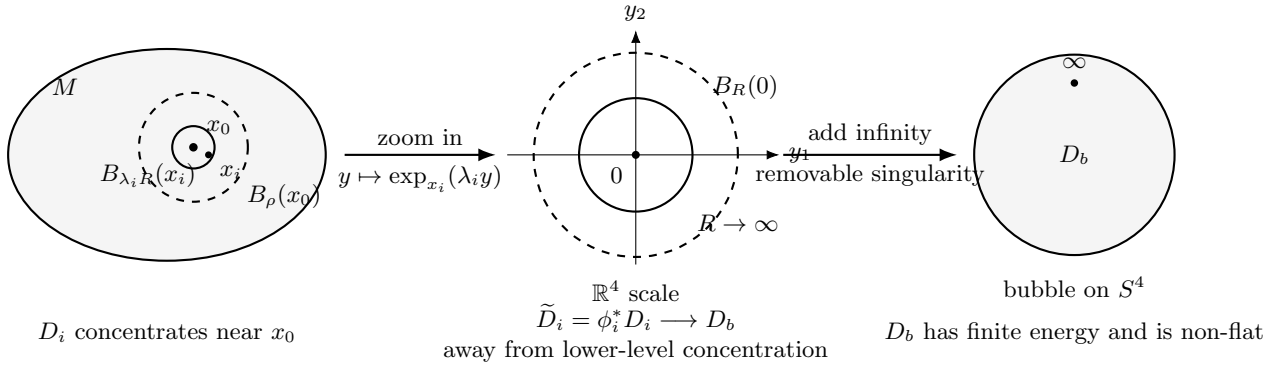


Figure 1: The basic blow-up picture. A small ball $B_{\lambda_i R}(x_i)$ near a concentration point is rescaled to a fixed ball $B_R(0)$ in $\mathbb{R}^4$. The pulled-back connections are Yang–Mills with respect to the rescaled metrics $g_i = \lambda_i^{-2} \phi_i^* g$, which converge smoothly to the Euclidean metric on compact subsets. After passing to gauges and subsequences, the rescaled connections converge away from possible lower-level concentration points to a finite-energy Yang–Mills connection on $\mathbb{R}^4$, which extends across infinity to a Yang–Mills connection on S^4 .

If the scale is chosen so that the curvature is normalized, for example $|F_{\tilde{D}_i}|_{g_i}(0) = 1$, then ε -regularity and compactness give, after passing to a subsequence and applying gauges, a smooth limit on compact subsets away from any lower-level concentration points:

$$\tilde{D}_i \longrightarrow D_b.$$

In a curvature-maximal rescaling, or after excluding the secondary concentration points in the rescaled picture, this convergence is smooth near the base point of the rescaling. The limit D_b is a finite-energy Yang–Mills connection on $\mathbb{R}^4$. It is non-flat by the normalization whenever the convergence is smooth near the normalized point. By the removable singularity theorem at infinity, D_b extends to a Yang–Mills connection on a bundle over S^4 . This limiting object is called a bubble.

Definition 5.1 (Bubble). *A bubble is a non-flat finite-energy Yang–Mills connection on $\mathbb{R}^4$, equivalently a non-flat Yang–Mills connection on a bundle over S^4 .*

The normalization forces the bubble to be non-flat when the chosen rescaling has no secondary concentration at the normalized point. By the energy gap theorem, therefore,

$$\text{YM}(D_b) \geq \varepsilon_{\text{gap}}.$$

Consequently, each bubble carries a definite positive amount of energy. Since the original sequence has total energy bounded by Λ , only finitely many bubbles can appear.

5.2 Bubble-tree extraction

There may be more than one bubble at a concentration point. After extracting the first bubble, one examines the remaining energy. If energy still concentrates at a scale much smaller or much larger than the first scale, rescale again. This produces a bubble tree.

A convenient way to organize scales is to say that two sequences of scales λ_i and μ_i are separated if

$$\frac{\lambda_i}{\mu_i} + \frac{\mu_i}{\lambda_i} + \frac{d(x_i, y_i)^2}{\lambda_i \mu_i} \longrightarrow \infty.$$

Separated scales correspond to different vertices in the bubble tree.

Theorem 5.2 (Bubble-tree compactness, schematic form). *Let D_i be Yang–Mills connections on a compact four-manifold M with $\text{YM}(D_i) \leq \Lambda$. After passing to a subsequence, there are:*

- *a smooth Yang–Mills limit D_∞ on $M \setminus \mathcal{S}$, which extends removably to a bundle $E_\infty \rightarrow M$;*
- *finitely many first-level concentration points $x_\alpha \in M$;*
- *finitely many non-flat bubbles $D_{\alpha j}$ on bundles over S^4 ;*

such that, away from the concentration points and away from the bubble points in each rescaling, the sequence converges smoothly in suitable gauges. Moreover, each extracted non-flat bubble carries at least ε_{gap} energy, and the extraction is arranged on disjoint or scale-separated regions. Hence the total number of bubbles is bounded in terms of $\Lambda/\varepsilon_{\text{gap}}$.

Proof sketch. The finite set of first-level concentration points follows from the energy threshold. Away from them, Theorem 3.5 gives smooth convergence. The limiting connection is initially defined on $M \setminus \mathcal{S}$ and then extends smoothly across $\mathcal{S}$ by the removable singularity theorem, possibly on a different bundle $E_\infty \rightarrow M$.

At a concentration point, choose a curvature or energy scale and rescale. The rescaled connections are Yang–Mills with respect to rescaled metrics converging smoothly to the Euclidean metric on compact subsets. Uhlenbeck compactness and ε -regularity give a smooth non-flat finite-energy limit on $\mathbb{R}^4$ away from any lower-level concentration points. By removable singularities, this limit extends across infinity, and across the lower-level points after further extraction, producing bubbles on S^4 . If energy remains between the base and this bubble, repeat the procedure at the remaining concentration scales. Each non-flat bubble consumes at least ε_{gap} energy, and the scale-separation construction prevents the same energy from being counted twice. Therefore the process terminates after finitely many bubbles. $\square$

5.3 Necks and the energy identity

The basic blow-up picture extracts the base connection and the bubbles. What remains is to understand the intermediate region between the base scale and the bubble scale. This intermediate region is called a neck. The key analytic fact is that, in dimension four, no curvature energy is lost in the neck.

Suppose a bubble forms near $x_i \rightarrow x_0$ at scale $\lambda_i \downarrow 0$. Fix a small number $\delta > 0$ and a large number $R > 1$. Then the region near x_i is decomposed into three pieces:

$$B_\delta(x_i) = (B_\delta(x_i) \setminus B_{R\lambda_i}(x_i)) \cup B_{R\lambda_i}(x_i).$$

The outer region belongs to the body, the annulus

$$A_i(\delta, R) := B_\delta(x_i) \setminus B_{R\lambda_i}(x_i)$$

is the neck, and the inner region is the bubble region after rescaling.

Under the logarithmic coordinate $t = -\log r$, the neck annulus becomes a long cylindrical region. More precisely, after shifting the t -coordinate, one may view it as $S^3 \times [T_i^-, T_i^+]$, where the cylinder length satisfies

$$T_i^+ - T_i^- \sim \log\left(\frac{\delta}{R\lambda_i}\right) \longrightarrow \infty.$$

Thus the neck is a long cylinder connecting the body to the bubble.

The analytic no-neck statement is that the curvature energy on the neck tends to zero:

$$\lim_{R \rightarrow \infty} \lim_{\delta \downarrow 0} \limsup_{i \rightarrow \infty} \int_{A_i(\delta, R)} |F_{D_i}|^2 = 0. \quad (16)$$

This is exactly the statement that all energy is accounted for by the base limit and the bubbles.

The reason for (16) is the same in spirit as in the cylinder proof of the removable singularity theorem, but one has to formulate the neck estimate carefully. After all bubbles between two adjacent scales have

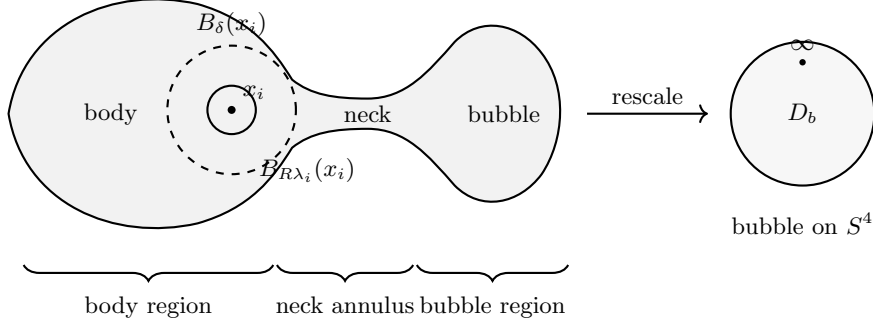


Figure 2: A schematic picture of the body–neck–bubble decomposition near a concentration point. The neck is the annular region $A_i(\delta, R) = B_\delta(x_i) \setminus B_{R\lambda_i}(x_i)$ connecting the body to the bubble scale. After rescaling the bubble region, one obtains a non-flat finite-energy Yang–Mills connection on S^4 .

been extracted, the remaining neck has no further concentration in the following quantitative sense: for every small $\varepsilon > 0$, after choosing the bubble-tree scales and then taking i sufficiently large,

$$\sup_{t \in [T_i^-, T_i^+ - 1]} \int_{S^3 \times [t, t+1]} |F_{D_i}|^2 < \varepsilon. \quad (17)$$

Thus every unit-length subcylinder has small energy, and by ε -regularity the curvature is uniformly small there.

Figure 3 gives a schematic picture of the energy distribution directly on the neck. The shaded ribbon inside the cylinder represents the slice energy. It is small near the two ends, and the neck estimate prevents a new concentration peak from forming in the middle. Thus the neck carries no independent limiting energy.

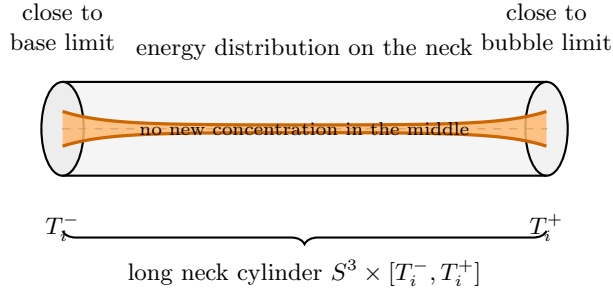


Figure 3: Schematic energy distribution on the neck. The shaded ribbon represents the slice energy along the long cylinder. It is small near the two ends and remains controlled in the interior by the neck estimate, so no independent neck energy survives in the limit.

To describe the neck estimate, write

$$f_i(t) := \int_{S^3 \times \{t\}} |F_{D_i}|^2.$$

One should not regard f_i as automatically satisfying a clean scalar inequality $f_i'' \geq c f_i$ directly from the Weitzenböck formula. The actual argument uses suitable small-curvature Coulomb or Hodge gauges on the cylinder, decomposes the curvature into tangential and radial components, and uses the spectral gap on S^3 to obtain a superconvexity or three-annulus estimate for the angular energy. Schematically, this estimate says that, on a long neck satisfying (17), the energy in the middle part of the cylinder is controlled by the energies near the two ends.

There is also a small technical point at the ends. The convergence to the base and to the bubble gives small energy on short collar regions near the two ends, not necessarily on the exact boundary slices. By the coarea formula one chooses good boundary slices inside these collars, say $S^3 \times \{\tau_i^-\}$ and $S^3 \times \{\tau_i^+\}$, whose slice energies are small. Applying the superconvexity or three-annulus estimate between these two

good slices gives

$$\int_{S^3 \times [\tau_i^-, \tau_i^+]} |F_{D_i}|^2 \leq C \left(\int_{\text{left collar}} |F_{D_i}|^2 + \int_{\text{right collar}} |F_{D_i}|^2 \right) + o_i(1),$$

in the schematic sense needed for the bubble-tree argument. The left collar is close to the base limit, so its energy becomes small once δ is chosen small. The right collar is close, after rescaling, to the bubble limit near infinity, so its energy becomes small once R is chosen large. Hence the whole neck energy tends to zero:

$$\int_{T_i^-}^{T_i^+} f_i(t) dt = \int_{\text{neck}_i} |F_{D_i}|^2 \longrightarrow 0.$$

This proves (16).

Theorem 5.3 (Energy identity). *Let D_i be a bubble-tree convergent sequence of Yang–Mills connections on a compact four-manifold, and assume the no-neck energy estimate (16) has been established for all necks in the bubble tree. Then*

$$\lim_{i \rightarrow \infty} \int_M |F_{D_i}|^2 = \int_M |F_{D_\infty}|^2 + \sum_{\alpha, j} \int_{S^4} |F_{D_{\alpha j}}|^2. \quad (18)$$

Proof sketch. Decompose M into three kinds of regions: the body region away from small balls around the concentration points, the bubble regions after rescaling, and the neck annuli between them. On the body region, smooth convergence gives convergence of the curvature energy to that of D_∞ . On the bubble regions, smooth convergence of the rescaled connections gives the energies of the bubbles $D_{\alpha j}$. The remaining energy lies in the neck annuli, and (16) shows that this contribution tends to zero. Summing over all bubbles gives (18). $\square$

There are also two related no-neck statements. The first is analytic: no L^2 -curvature energy is lost in the necks, which is precisely the content of the energy identity. The second is topological: the transition data along a neck is homotopically constant after the base and bubble pieces have been fixed, so the original bundle is obtained by gluing, or connected summing, the base bundle with the bubble bundles. Thus a neck carries neither extra energy nor new topology; it only records how the body and the bubbles are attached.

5.4 Topology carried by bubbles

The energy identity is the analytic no-neck statement: no L^2 -curvature energy remains on the necks. There is also a topological no-neck statement. Although all connections D_i are defined on the same bundle $E \rightarrow M$, the smooth limit away from the concentration points extends, after removable singularities, to a bundle $E_\infty \rightarrow M$ which need not be isomorphic to E . The topology lost from the base is recovered by the bubble bundles over S^4 .

For example, when $G = \text{SU}(2)$, the second Chern number is represented by a Chern–Weil integral. With a fixed trace convention, one may write

$$c_2(E) = \frac{1}{8\pi^2} \int_M \text{tr}(F_D \wedge F_D),$$

up to the usual possible sign depending on the convention for tr and for the Lie algebra. Under bubbling, the corresponding characteristic number splits as

$$c_2(E) = c_2(E_\infty) + \sum_{\alpha, j} c_2(E_{\alpha j}).$$

This splitting is topological. It is related to, but logically distinct from, the energy identity.

We now describe the connected-sum picture for bundles. Let $E_0 \rightarrow M$ be a bundle and let $E_b \rightarrow S^4$ be a bubble bundle. Remove a small ball from M and a small ball from S^4 . The two new boundary components are both copies of S^3 . After choosing boundary trivializations, one can glue the two restricted bundles along these boundary spheres. The resulting bundle is denoted schematically by

$$E_0 \# E_b \longrightarrow M \# S^4 \cong M.$$

This notation includes the choice of boundary homotopy class. Indeed, changing a trivialization over S^3 changes the clutching map by a map $S^3 \rightarrow G$, and this may be nontrivial even when G is connected. In the bubbling situation, the neck determines the boundary identification between the base and bubble pieces. The topological no-neck statement says that this identification contributes no additional homotopy class beyond those already carried by the base and bubble bundles.

Theorem 5.4 (Topological identity, schematic form). *Let D_i be a bubble-tree convergent sequence of Yang–Mills connections on a fixed G -bundle $E \rightarrow M$, where G is compact and connected. Let $E_\infty \rightarrow M$ be the removable extension of the base limiting bundle, and let $E_{\alpha j} \rightarrow S^4$ be the bubble bundles. Then the original bundle is obtained topologically by connected summing the base bundle with the bubble bundles:*

$$E \cong E_\infty \# (\#_{\alpha, j} E_{\alpha j}). \quad (19)$$

The connected-sum operation in (19) is understood with the boundary identifications determined by the neck regions. The topological no-neck statement says that these identifications contribute no independent neck class. Consequently, degree-four Chern–Weil characteristic numbers split additively.

Proof. It is enough to explain the mechanism near one bubble; the full bubble tree is obtained by repeating the same argument on each neck. Remove from the base a small ball around the concentration point, and remove from the bubble a small ball around the point corresponding to infinity. The remaining intermediate region is a neck, conformally a long cylinder $S^3 \times [a_i, b_i]$.

On the outer side of the neck, smooth convergence identifies the original bundle with the base limiting bundle. On the inner side, after rescaling, smooth convergence identifies the original bundle with the bubble bundle. Thus the only point to check is that the two boundary trivializations are compatible through the neck without producing an additional homotopy class.

By construction there is no further concentration on the neck. Hence the curvature is small on every unit-length subcylinder. The same small-curvature gauge construction used in the removable singularity theorem gives trivializations on the slices $S^3 \times \{t\}$ in which the connection one-form is small. If two such slice trivializations are compared, their transition map $g : S^3 \rightarrow G$ satisfies an estimate of the form

$$\|dg\|_{W^{1,p}(S^3)} \leq C\varepsilon$$

for some $p > 2$, after choosing the neck energy threshold sufficiently small. After fixing a constant element $g_0 \in G$, a Poincaré estimate gives

$$\|g - g_0\|_{W^{2,p}(S^3)} \leq C\varepsilon.$$

Since $p > 2$ and S^3 is three-dimensional, the Sobolev embedding $W^{2,p}(S^3) \hookrightarrow C^0(S^3)$ implies that g has small oscillation. Thus the image of g lies in a geodesic ball in G , and so g is homotopic to a constant map. Transporting this comparison along the long cylinder shows that the outer and inner boundary framings determine the same clutching data up to the classes already carried by the base and bubble pieces.

Thus the neck contributes no independent topological class. All nontrivial topology is carried either by the removable base limit or by the bubble bundles. Cutting along the necks and gluing the base and bubble pieces back together gives (19). The additivity of characteristic numbers follows by applying Chern–Weil theory to this decomposition, or equivalently by observing that the neck contribution tends to zero while the base and bubble regions converge smoothly. $\square$

6 Main theorem in final form

Theorem 6.1 (Yang–Mills blow-up theorem, four-dimensional form). *Let (M^4, g) be a compact Riemannian four-manifold, and let $E \rightarrow M$ be a smooth bundle with compact structure group G . Let D_i be a sequence of smooth Yang–Mills connections on E such that*

$$\int_M |F_{D_i}|^2 \leq \Lambda.$$

Then, after passing to a subsequence, there exist a finite set $\mathcal{S} = \{x_1, \dots, x_N\} \subset M$, a smooth Yang–Mills connection D_∞ on a smooth bundle $E_\infty \rightarrow M$, and finitely many non-flat Yang–Mills connections $D_{\alpha j}$ on bundles $E_{\alpha j} \rightarrow S^4$, such that the following hold.

First, after gauge transformations, D_i converges smoothly to D_∞ on compact subsets of $M \setminus \mathcal{S}$. Secondly, after choosing appropriate centers and scales near each x_α , the rescaled connections converge smoothly on compact subsets of $\mathbb{R}^4$ away from lower-level concentration points, and each rescaled limit extends across infinity to one of the bubbles $D_{\alpha j}$ on S^4 .

Moreover, the energy identity holds:

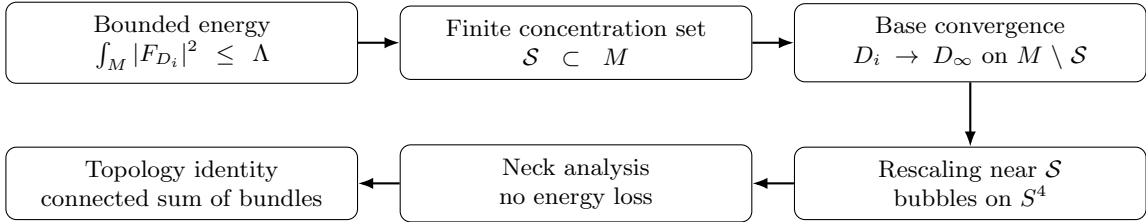
$$\lim_{i \rightarrow \infty} \text{YM}(D_i) = \text{YM}(D_\infty) + \sum_{\alpha, j} \text{YM}(D_{\alpha j}). \quad (20)$$

Finally, the original bundle is recovered topologically by connected summing the base limiting bundle with the bubble bundles:

$$E \cong E_\infty \# (\#_{\alpha, j} E_{\alpha j}), \quad (21)$$

where the connected-sum data are the boundary homotopy classes determined by the neck regions.

Proof. The proof is the bubble-tree argument assembled from the preceding analytic inputs. The logical structure is as follows.



The first step is to locate the concentration set. Choose the small constant from the ε -regularity theorem and define $\mathcal{S}$ to be the set of points at which at least this amount of energy persists at arbitrarily small scales. Since the total energy is bounded by Λ , the set $\mathcal{S}$ is finite.

On compact subsets of $M \setminus \mathcal{S}$, the curvature energy is uniformly small on sufficiently small balls. Uhlenbeck gauge fixing puts the connections in Coulomb gauges, and the Yang–Mills equation becomes an elliptic system in these gauges. The ε -regularity theorem then gives uniform higher-order estimates. Passing to a subsequence gives smooth convergence away from $\mathcal{S}$. The removable singularity theorem then extends the base limit smoothly across each point of $\mathcal{S}$, possibly on a different bundle $E_\infty \rightarrow M$.

The second step is to zoom in near each point of $\mathcal{S}$. Choosing centers and scales where a definite amount of energy concentrates, one obtains rescaled Yang–Mills connections on larger and larger subsets of $\mathbb{R}^4$. The same compactness argument gives a nontrivial finite-energy Yang–Mills limit on $\mathbb{R}^4$ away from secondary concentration points. By removable singularities, this limit extends across infinity and across the secondary points, producing bubbles on S^4 . Repeating the construction gives the bubble tree.

The bubble tree is finite because of the energy gap on S^4 : every non-flat bubble carries at least a fixed positive amount of energy. Since the original energy is bounded by Λ , only finitely many such bubbles can appear.

The third step is the no-neck statement. Between two adjacent scales, the annular region is conformal to a long cylinder. By construction, there is no further concentration on this cylinder, so the curvature is small on every unit-length subcylinder. The cylinder estimate proved earlier implies that the energy on the neck tends to zero:

$$\lim_{i \rightarrow \infty} \int_{\text{neck}_i} |F_{D_i}|^2 = 0.$$

Thus all energy is accounted for by the base limit and the bubbles, which gives (20).

Finally, the topological identity follows from the corresponding no-neck statement for boundary trivializations. On each neck, small-curvature gauges and parallel transport identify the trivializations at the two ends up to homotopy. Hence the neck contributes no independent topological class. The original bundle is therefore obtained by gluing together the base limiting bundle and the bubble bundles along the boundary S^3 's, giving (21). $\square$

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